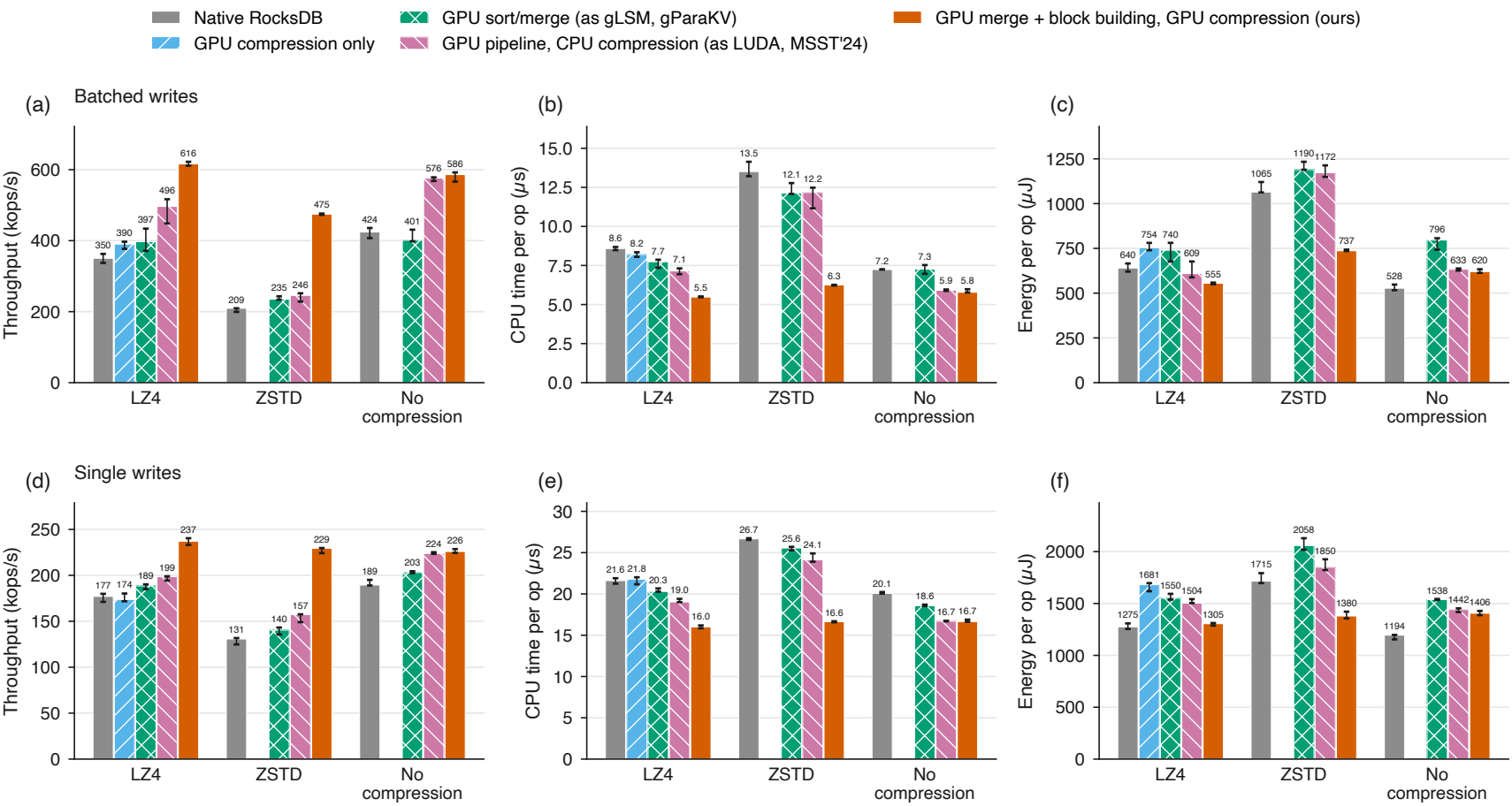
Native RocksDB
GPU compression only
GPU sort/merge (as gLSM, gParaKV)
GPU pipeline, CPU compression (as LUDA, MSST'24)
GPU merge + block building, GPU compression (ours)
(a)
Batched writes
Throughput (kops/s)
0
200
400
600
350
390
397
496
616
209
235
246
475
424
401
576
586
LZ4
ZSTD
No compression
(b)
CPU time per op (μs)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
8.6
8.2
7.7
7.1
5.5
13.5
12.1
12.2
6.3
7.2
7.3
5.9
5.8
LZ4
ZSTD
No compression
(c)
Energy per op (μJ)
0
250
500
750
1000
1250
640
754
740
609
555
1065
1190
1172
737
528
796
633
620
LZ4
ZSTD
No compression
(d)
Single writes
Throughput (kops/s)
0
50
100
150
200
250
177
174
189
199
237
131
140
157
229
189
203
224
226
LZ4
ZSTD
No compression
(e)
CPU time per op (μs)
0
5
10
15
20
25
30
21.6
21.8
20.3
19.0
16.0
26.7
25.6
24.1
16.6
20.1
18.6
16.7
16.7
LZ4
ZSTD
No compression
(f)
Energy per op (μJ)
0
500
1000
1500
2000
1275
1681
1550
1504
1305
1715
2058
1850
1380
1194
1538
1442
1406
LZ4
ZSTD
No compression